

Section : Physics

Q.1 The momentum of a photon associated with a radiation of frequency 2×10^{13} Hz is
(Given: $h = 6.6 \times 10^{-34}$ J s and $c = 3 \times 10^8$ m/s)

1. 1.1×10^{-29} kg m/s
2. 2.2×10^{-29} kg m/s
3. 1.1×10^{-28} kg m/s
4. 4.4×10^{-29} kg m/s

Options 1. 1
2. 2
3. 3
4. 4

Q.2 Which of the following are explained in Rutherford's model of an atom?
(A) An atom consists of a core called nucleus.
(B) The nucleus occupies a very small space as compared to the size of atom.
(C) The stability of an atom.
(D) The electrons revolve around the nucleus in various orbits.
Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (D) only
3. (B), (C) and (D) only
4. (A), (B) and (C) only

Options 1. 1
2. 2
3. 3
4. 4

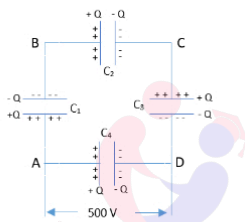
Q.3 A 10 V battery of negligible internal resistance is being charged by a 200 V battery with a resistance of 38Ω in series. The value of the current in the circuit is

1. 5.5 A
2. 5 A
3. 5.26 A
4. 0.26 A

Options 1. 1
2. 2
3. 3
4. 4

Q.4

A network of four $9\mu\text{F}$ capacitors is connected to a 500 V supply, as shown. In the context of this, identify the correct statements.



- (A) C_4 is connected in parallel to the combination of C_1 , C_2 and C_3 .
- (B) The equivalent capacitance of the combination C_1 , C_2 , C_3 and C_4 is $12\mu\text{F}$.
- (C) The charge on C_4 is 4.5×10^{-3} C.
- (D) The charge stored on C_1 and C_4 is the same.

Choose the **correct** answer from the options given below:

1. (B), (C) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (A), (B), (C) and (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.5 In Kirchhoff's loop rule

1. The currents entering the junction are not equal to the sum of currents leaving the junction.

2. In a circuit, charge is accumulated at a point.

3. In a closed circuit, the potential difference across the cell is equal to the electromotive force of the cell.

4. The algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Two concentric coils X and Y , each of radius 2π cm, are placed at right angles to each other. If X and Y carry 4 A and 3 A currents, respectively, then the value of the magnetic field (in Wb m^{-2}) at the centre of the coils is

(Take $\mu_0 = 4\pi \times 10^{-7} \text{ Wb A}^{-1}\text{m}^{-1}$)

1. 1.7×10^{-5}

2. 2.6×10^{-5}

3. zero

4. 4.5×10^{-5}

Options 1. 1

2. 2

3. 3

4. 4

Q.7 When current flowing in an inductive circuit is switched off, the induced current will be

1. in the opposite direction as the main current

2. equal to the main current

3. in the same direction as the main current

4. zero

Options 1. 1

2. 2

3. 3

4. 4

Q.8 To obtain a n-type silicon semiconductor, it must be doped with

1. Aluminium

2. Phosphorus

3. Boron

4. Magnesium

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Match List-I with List-II

List - 1

Observation

(A) Blue colour of sky

(B) Seven colours obtained from prism

(C) Bending of pencil dipped in water

(D) Alternate bright and dark bands of equal thickness

List - 2

Light related phenomenon

(I) Dispersion

(II) Refraction

(III) Scattering

(IV) Interference

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

2. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)

3. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
 4. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

Options 1. 1

2. 2
 3. 3
 4. 4

- Q.10** A point object is placed at a distance of 20 cm from a spherical lens of refractive index $n=3/2$ and the radius of curvature of each surface 10 cm. The distance of the image formed from the lens is
 1. 1.5 cm
 2. 10 cm
 3. 20 cm
 4. 25 cm

Options 1. 1

2. 2
 3. 3
 4. 4

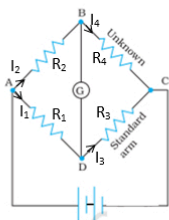
- Q.11** A circular coil of area 300 cm^2 and 25 turns rotates about its vertical diameter with an angular speed of 40 rad s^{-1} in a uniform horizontal magnetic field of magnitude 0.05 T. The maximum voltage induced in the coil is
 1. 1.1 V
 2. 1.5 V
 3. 2.5 V
 4. 2.8 V

Options 1. 1

2. 2
 3. 3
 4. 4

Q.12

In a Wheatstone bridge, the resistances R_1 , R_2 , R_3 and R_4 present in four arms are 10Ω , 30Ω , 30Ω and 90Ω respectively. If the galvanometer resistance is 50Ω and the voltage of the battery is 7 V and its internal resistance is 5Ω , then the total current drawn from the battery is



1. 1.2 A
 2. 0.1 A
 3. 2.0 A
 4. 0.2 A

Options 1. 1

2. 2
 3. 3
 4. 4

Q.13 Match List-I with List-II

List- I

- (A) For a series LC circuit
 (B) For a parallel LC circuit
 (C) For a purely capacitive circuit
 (D) For a series LCR circuit at resonance

List- II

- (I) Currents in L and C are 180° out of phase
 (II) Power Factor = 0
 (III) Voltage across L and C are 180° out of phase
 (IV) Reactance is zero

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
 2. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
 3. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
 4. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Two identical conducting wires, A and B of different materials, are joined in series across a battery. If the number density of electrons in A is thrice that in B, then the ratio of drift velocity of electrons in the two wires is

1. 1:1

2. 1:2

3. 1:3

4. 1:4

Options 1. 1

2. 2

3. 3

4. 4

Q.15 The shortest wavelength in the Balmer series of a hydrogen atom is
(Given: $R = 1.1 \times 10^7 \text{ m}^{-1}$)

1. 9116 \AA

2. 6562 \AA

3. 3637 \AA

4. 8204.1 \AA

Options 1. 1

2. 2

3. 3

4. 4

Q.16 There are four particles A, B, C and D with the same mass and equal kinetic energy K. If the kinetic energy of particle A becomes 4 K, B becomes 9 K, C becomes 64 K and D becomes 16 K times, respectively, the fractional change in the de Broglie wavelength of the particles arranged in increasing order is

1. (A), (B), (D), (C)

2. (C), (B), (D), (A)

3. (B), (A), (D), (C)

4. (C), (D), (B), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Match List-I with List-II

List-I

List-II

(A) Remote control of a TV (I) Radio waves

(B) Radar (II) Infrared Waves

(C) Sonar (III) Microwaves

(D) Mobile phones (IV) Ultrasonic Waves

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.18 A non-polar dielectric material is placed in an external electric field. Choose the correct statement(s) from the following:
(A) There will be no change in the positions of positive and negative charges inside the dielectric.
(B) The positive charges of molecules will shift opposite to the direction of the electric field.

(C) The negative charges of molecules will shift in the direction of the electric field.
(D) The positive charges of the molecules will shift in the direction of the field and negative charges of the molecules will shift opposite to the field.

Choose the correct answer from the options given below:

1. (D) only
2. (B) and (C) only
3. (A) only
4. (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.19 A solenoid has a core of a material with a relative permeability of 500. The windings of the solenoid are insulated from the core and carry a current of 0.5 A. If the number of turns is 2000 per metre, the magnetic intensity is

1. $4 \times 10^3 \text{ A m}^{-1}$
2. 10^3 A m^{-1}
3. $0.2\pi \text{ A m}^{-1}$
4. $0.2\pi \text{ A m}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.20 Forward bias applied to a p-n junction diode

1. widens the depletion region
2. increases the number of donors on the n-side
3. decreases the electric field in the depletion region
4. increases the potential difference across the depletion region

Options 1. 1

2. 2
3. 3
4. 4

Q.21 The storage battery of a car has an emf of 10 V. If the internal resistance of the battery is 0.2Ω , then the maximum current that can be drawn from the battery is

1. 100 A
2. 10 A
3. 5 A
4. 50 A

Options 1. 1

2. 2
3. 3
4. 4

Q.22 In an a. c. series LCR circuit, the applied voltage is $v = (100\sqrt{2}) \sin(\omega t + \pi/4)$ volt. Given that $R = 30\Omega$, $X_L = 50\Omega$ and $X_C = 10\Omega$. The value of power dissipated in the circuit is

1. 200 W
2. Zero
3. 100 W
4. 120 W

Options 1. 1

2. 2
3. 3
4. 4

Q.23 At one place on earth, the resultant magnetic field is 0.25 G and the angle of dip is 45° . The horizontal component of earth's magnetic field will be

1. 0.18 G
2. 0.25 G

3. 0.35 G
4. 0.125 G

Options 1. 1
2. 2
3. 3
4. 4

Q.24 A long solenoid has 500 turns. When a current of 2 A is passed through it, the resulting magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self-inductance of the solenoid is

1. 4 H
2. 3 H
3. 2 H
4. 1 H

Options 1. 1
2. 2
3. 3
4. 4

Q.25 In a galvanometer, there is a deflection of 10 divisions per mA. The resistance of the galvanometer is 60Ω . If a shunt of 1Ω is connected to the galvanometer and there are 100 divisions in all, on the scale of the galvanometer, the maximum current that this galvanometer can read is

1. 600 mA
2. 610 mA
3. 10 mA
4. 100 mA

Options 1. 1
2. 2
3. 3
4. 4

Q.26 A rectangular coil of sides 2 cm and 5 cm having 2000 turns and carrying a current of 200 mA is placed in a uniform magnetic field of 5.0 T directed along positive X-axis. The maximum torque experienced by the coil is

1. 2N m
2. 4N m
3. 0.2 N m
4. 0.4 N m

Options 1. 1
2. 2
3. 3
4. 4

Q.27 The focal length of a converging lens is measured for violet, green, red and yellow colours. Arrange the focal length for different colours in increasing order.

- (A) fv
(B) fg
(C) fr
(D) fy

Choose the correct answer from the options given below:

1. (B), (A), (C), (D)
2. (A), (B), (D), (C)
3. (D), (B), (C), (A)
4. (A), (B), (C), (D)

Options 1. 1
2. 2
3. 3
4. 4

Q.28 Which of the following is NOT the property of electric field lines?

1. They start from positive charges and end at negative charges.
2. They can never cross each other.

3. They form closed loops.
4. They are continuous curves in a charge-free region.

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Match List-I with List-II

List -I	List II
Physical Quantity	Dimensional Formula
(A) Capacitance	(I) $[ML^3T^{-3}A^{-1}]$
(B) Electric Flux	(II) $[ML^2T^{-3}A^{-1}]$
(C) Potential	(III) $[L^{-2}AT]$
(D) Polarisation	(IV) $[M^{-1}L^{-2}T^{-4}A^2]$

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.30 A nucleus ${}_{92}^{238}\text{X}$ emits one alpha particle and then one beta particle (electron). The product nucleus will have

1. Atomic number = 90, Mass number = 234
2. Atomic number = 91, Mass number = 234
3. Atomic number = 91, Mass number = 238
4. Atomic number = 89, Mass number = 234

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The electric field in a certain region is acting radially outward and is given by $E = Ar^2$. A charge contained in a sphere of radius 'a' centred at the origin of the field, will be given by

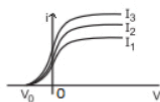
1. $4\pi\epsilon_0 Aa^3$
2. $\pi\epsilon_0 Aa^3$
3. $4\pi\epsilon_0 Aa^4$
4. $\epsilon_0 Aa^3$

Options 1. 1

2. 2
3. 3
4. 4

Q.32

Figure given below shows variation of photocurrent with anode potential for different intensities of incident radiation. Some of the results obtained from the experiments are described below on the basis of the graph. Identify the correct results:



- (A) The photoelectric current is higher for higher intensity of the incident radiation.
- (B) The negative potential required to decrease the photocurrent at different intensities to zero is different.
- (C) If the positive potential of the anode is increased beyond a certain value, the photocurrent does not increase for a given intensity.
- (D) If the potential applied at the anode is zero, then the photocurrent is lower than the maximum value of current for the given intensity.

Choose the correct answer from the options given below:

- 1. (A), (B) and (C) only
- 2. (B), (C) and (D) only
- 3. (A), (C) and (D) only
- 4. (A) and (C) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.33 Motion of a charged particle having velocity v in a uniform magnetic field B is circular, if v is

- 1. parallel to B
- 2. anti-parallel to B
- 3. perpendicular to B
- 4. at any angle to B

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.34 The power and aperture of lenses L1, L2 and L3 are given below in tabular format. Which pair of lenses should be chosen for constructing a microscope?

Lens Power Aperture

L1 3D 8 cm

L2 6D 1 cm

L3 10D 1 cm

- 1. L2 as eyepiece and L3 as objective
- 2. L3 as eyepiece and L2 as objective
- 3. L2 as eyepiece and L1 as objective
- 4. L1 as eyepiece and L2 as objective

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.35 Which of the following examples are based upon the total internal reflection of light?

- (A) Optical fibre
- (B) Twinkling of stars
- (C) Mirage
- (D) Sparkling of diamond

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (C) and (D) only
- 3. (B), (C) and (D) only
- 4. (A), (B) and (C) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.36 Two positive point charges of $q_1 = 0.2\mu\text{C}$ and $q_2 = 0.01\mu\text{C}$ are placed $r_i = 10\text{ cm}$ apart. The work done in reducing the distance to $r_f = 5\text{ cm}$ will be

1. $1.8 \times 10^{-4}\text{ J}$
2. $1.8 \times 10^{-5}\text{ J}$
3. 0.018 J
4. $1.8 \times 10^{-3}\text{ J}$

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Match List-I with List-II

List-I

List-II

Physical Quantity

SI UNIT

(A) Mobility of a charge carrier (I) $\Omega^{-1}\text{ m}^{-1}$

(B) Resistivity (II) A m^{-2}

(C) Current Density (III) $\text{m}^2\text{ V}^{-1}\text{ s}^{-1}$

(D) Conductivity (IV) $\Omega\text{ m}$

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.38 A plano-convex lens has focal length 0.5 m and the refractive index of the material of the lens is 1.5 . The radius of curvature of the convex surface of the lens is

1. 0.15 m
2. 0.30 m
3. 0.25 m
4. 0.50 m

Options 1. 1

2. 2
3. 3
4. 4

Q.39 The separation between the plates of a parallel plate capacitor of capacitance 2 pF is 0.5 mm . The area of the plates of the capacitor is

(Given: $\epsilon_0 = 8.85 \times 10^{-12}\text{ C}^2\text{ N}^{-1}\text{ m}^{-2}$)

1. 520 cm^2
2. 71 cm^2
3. 113 cm^2
4. 1.13 cm^2

Options 1. 1

2. 2
3. 3
4. 4

Q.40 A small telescope has an objective lens of focal length 120 cm and an eyepiece of focal length of 5.0 cm. The magnifying power of telescope for viewing distant objects in normal adjustment is

1. 1.28
2. 2.24
3. 3.20
4. 4.22

Options

1. 1
2. 2
3. 3
4. 4

Q.41 If the temperature of a metallic conductor is increased, then the product of resistivity and conductivity

1. increases linearly with temperature
2. decreases
3. remains same
4. proportional to square of the temperature

Options

1. 1
2. 2
3. 3
4. 4

Q.42 A conducting circular loop is placed in a uniform magnetic field of 10 T with its plane normal to the field. Now, the radius of the loop starts shrinking at the rate 5 mm/s. When the radius is 5 cm then the induced emf is

1. 7.85 mV
2. 15.7 mV
3. 78.5 mV
4. 1.57 mV

Options

1. 1
2. 2
3. 3
4. 4

Q.43 Magnetic field at large distances on the axis of a current carrying circular loop depends on distance from the centre as

1. directly proportional to distance
2. inversely proportional to the distance
3. inversely proportional to the square of distance
4. inversely proportional to the cube of distance

Options

1. 1
2. 2
3. 3
4. 4

Q.44 Which of the following statements are true about electromagnetic waves?

(A) Accelerated charges radiate electromagnetic waves.

(B) The frequency of the electromagnetic wave equals the frequency of oscillation of the charge.

(C) No material medium is involved in the vibrations of the electric and magnetic fields.

(D) Electromagnetic waves of different wavelengths have different velocities in free space/vacuum.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (A), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.45 Arrange the following electromagnetic waves in the increasing order of their wavelength.

(A) X-rays

(B) Radio waves

(C) Gamma rays

(D) Microwaves

Choose the correct answer from the options given below:

1. (C), (B), (A), (D)
2. (C), (A), (D), (B)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Match List-I with List-II

List-I

List-II

(A) Faraday's Law

(I) Current induced in a conductor moving perpendicular to a magnetic field

(B) Lenz's Law

(II) Direction of force on a straight current carrying conductor placed perpendicular to a magnetic field

(C) Fleming's right-hand rule

(III) Magnitude of induced emf

(D) Fleming's left-hand rule

(IV) Direction of induced current

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
4. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.47 The binding energy per nucleon for ^{56}Fe is

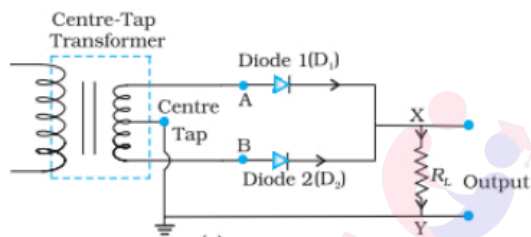
1. 8.75 MeV
2. 11.26 MeV
3. 9.96 MeV
4. 4.89 MeV

Options 1. 1

2. 2
3. 3
4. 4

Q.48

Arrange the following steps for working of a full wave rectifier in correct sequence



- (A) During the negative half cycle, D_1 is reverse biased while D_2 is forward biased.
- (B) During the positive half cycle, D_1 is forward biased while D_2 is reverse biased.
- (C) A pulsating d. c. voltage is obtained across R_L
- (D) a. c. input is given to the primary of a centre-tap transformer

Choose the **correct** answer from the options given below:

- 1. (D), (B), (C), (A)
- 2. (A), (D), (C), (B)
- 3. (D), (A), (C), (B)
- 4. (D), (B), (A), (C)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.49 Electric field at a point due to a uniformly charged infinite plane sheet is

- 1. proportional to distance from the sheet
- 2. proportional to square of the distance from the sheet
- 3. inversely proportional to square of the distance from the sheet.
- 4. independent of the distance from the sheet

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.50 In a single slit diffraction experiment, the aperture of the slit is 3 mm. A monochromatic light of wavelength 620 nm is normally incident on the slit. If the distance between the slit and the screen is 1.2 m, then the separation between the first order minima and the third order maxima on one side of the screen is

- 1. 1.620×10^{-4} m
- 2. 2.770×10^{-4} m
- 3. 6.20×10^{-4} m
- 4. 7.70×10^{-4} m

Options 1. 1

- 2. 2
- 3. 3
- 4. 4